# Intro

Five JDK tools for looking at the heap

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# Why look at the heap?

Five JDK tools show what is on the heap, how it grows and how big each object is.

- Garbage is invisible in the code, but obvious in a class histogram.
- Fewer and smaller objects mean fewer GCs and more headroom for a spike.
- Object size depends on the JVM: compact object headers shrink most small objects.
- The tools: `jmap`, `jcmd`, `jps`, `jstat` and Java Object Layout (JOL).

# Agenda

Five tools, from finding a running JVM to the exact layout of one object.

| # | Tool | In one line |
|---|---|---|
| 01 | jmap | Class histogram and heap dump of a running JVM |
| 02 | jcmd | One command for most diagnostics, including the histogram |
| 03 | jps | Finds the pid of every running JVM |
| 04 | jstat | GC and heap counters sampled over time |
| 05 | JOL | Header, fields and padding of one object |